Homework 08

REAL VARIABLES I

Fall 2026

Due: 2026-11-01 19:00
Topics: Hölder's inequality with several factors, sharpness of exponents, testing property of L^p norms, log-convexity of L^p norms, density of C_c in $L^p(\mathbb{R}^d)$.

Instructions

Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

In all problems, $(X, \mathcal{E}, \mu)$ is a measure space, f, g, f_j are measurable functions, and $\|\cdot\|_{L^p}$ is allowed to take the value $+\infty$; inequalities are understood in $[0, +\infty]$. For $p \in (1, \infty)$, p' denotes the conjugate exponent, $\frac{1}{p} + \frac{1}{p'} = 1$.

1. (Hölder's inequality with several factors)

1.a

weight: 1.0 Let $N \geq 1$ and $r, p_1, \dots, p_N \in (1, \infty)$ with $\frac{1}{r} = \sum_{j=1}^N \frac{1}{p_j}$. Prove that

$$\Big\| \prod_{j=1}^N f_j \Big\|_{L^r(X,\mu)} \leq \prod_{j=1}^N \|f_j\|_{L^{p_j}(X,\mu)}.$$

Free hint.

- You may use Hölder's inequality with two factors and exponents $p, p' \in (1, \infty)$.
- Induct on N, or reduce to the case $r = 1$ by applying that case to $|f_j|^r$.

- Treat separately the case where some $\|f_j\|_{L^{p_j}}$ is 0 or $+\infty$.

2. (Log-convexity of L^p norms)

2.a

weight: 1.0 Let $1 \le p_0 \le p_1 \le \infty$, $\theta \in [0,1]$, and $\frac{1}{p_\theta} \coloneqq \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$. Prove that

$$\|f\|_{L^{p_\theta}(X,\mu)} \le \|f\|^{1-\theta}_{L^{p_0}(X,\mu)} \|f\|^{\theta}_{L^{p_1}(X,\mu)}.$$

3. (Sharpness of Hölder's inequality)

Let $(X, \mathcal{E}, \mu)$ be a measure space with a set of positive finite measure, and let $p, q, r \in [1, \infty)$. Consider the inequality

$$\|fg\|_{L^r(X,\mu)} \le C \|f\|_{L^p(X,\mu)} \|g\|_{L^q(X,\mu)} \qquad (*)$$

for all measurable f, g, with a constant $C \in [0, \infty)$ independent of f, g.

3.a

weight: 0.5 Suppose $(*)$ holds.

- If $\frac{1}{r} > \frac{1}{p} + \frac{1}{q}$, prove that $\sup\{\mu(E) : E \in \mathcal{E},\ \mu(E) < \infty\} < \infty$, and determine the optimal value of C in terms of this supremum and p, q, r.
- If $\frac{1}{r} < \frac{1}{p} + \frac{1}{q}$, prove that $\inf\{\mu(E) : E \in \mathcal{E},\ \mu(E) > 0\} > 0$, and determine the optimal value of C in terms of this infimum and p, q, r.

3.b

weight: 0.3

- Conversely, suppose $\mu(X) < \infty$ (so the supremum in (a) equals $\mu(X)$). Prove that $(*)$ holds for any $\frac{1}{r} \ge \frac{1}{p} + \frac{1}{q}$ with the optimal constant you determined above .
- Conversely, suppose the infimum in (a) is positive. Prove that $(*)$ holds for any $\frac{1}{r} \le \frac{1}{p} + \frac{1}{q}$ with the optimal constant you determined above.

3.c

weight: 0.2 Give examples of a space and two functions and exponents $p, q, r \in [1, \infty)$ such that

$$\|fg\|_{L^r} = +\infty, \qquad \|f\|_{L^p} = 1, \|f\|_{L^q} = 1$$

with $\frac{1}{r} > \frac{1}{p} + \frac{1}{q}$. Separately given an example for which

$$\|fg\|_{L^r} = +\infty, \qquad \|f\|_{L^p} = 1, \|f\|_{L^q} = 1$$

with $\frac{1}{r} < \frac{1}{p} + \frac{1}{q}$.

4. (Testing property of L^p norms)

(In class.)

Let $(X, \mathcal{E}, \mu)$ be σ-finite, $p \in (1, \infty)$, and let $f\colon X \to \mathbb{R}$ be measurable.

4.a

weight: 0.6 Prove that

$$\|f\|_{L^p(X,\mu)} = \sup\Big\{ \int_X \Big| fg \,\mathrm{d}\mu \Big| \colon \|g\|_{L^{p'}(X,\mu)} < 1 \Big\}.$$

4.b

weight: 0.4 Prove that

$$\|f\|_{L^p(X,\mu)} = \sup\Big\{ \Big| \int_X fg \,\mathrm{d}\mu \Big| \colon \|g\|_{L^{p'}(X,\mu)} < 1,\ g \text{ simple} \Big\}.$$

Free hint.

- Make sure to address the case $\|f\|_{L^p} = +\infty$: you must show that the supremum is then also $+\infty$.
- For $\|f\|_{L^p} = \infty$, truncate: replace f by $f\mathbb{1}_{X_n \cap \{|f| \le n\}}$ with $X_n \uparrow X$ of finite measure.
- For (b), approximate the test function from (a) by simple functions; control the error.

5. (Density of compactly supported continuous functions)

Let μ be a Borel measure on $\mathbb{R}^d$ with $\mu(K) < +\infty$ for every compact set $K \subseteq \mathbb{R}^d$, and let $p \in (0, +\infty)$. Denote by $C_c(\mathbb{R}^d)$ the continuous functions $g\colon \mathbb{R}^d \to \mathbb{R}$ whose support $\operatorname{spt} g = \overline{\{g \neq 0\}}$ is compact, and write $B_r \coloneqq \overline{B}(0, r)$. You may use the results of Notes 27: density of simple functions in L^p, regularity of finite Borel measures on $\mathbb{R}^d$ (with compact inner approximations), and the Urysohn functions $\mathbb{1}_F \le \varphi \le \mathbb{1}_U$ for F closed and U open with $F \subseteq U$.

5.a

weight: 0.2 Prove that $C_c(\mathbb{R}^d) \subseteq L^p(\mathbb{R}^d, \mu)$, and that for every $f \in L^p(\mathbb{R}^d, \mu)$ we have $\|f - f\mathbb{1}_{B_R}\|_{L^p} \to 0$ as $R \to \infty$.

5.b

weight: 0.4 Let $R > 0$, let $E \subseteq B_R$ be a Borel set, and let $\eta > 0$. Prove that there exists $\varphi \in C_c(\mathbb{R}^d)$ with $0 \le \varphi \le 1$, $\operatorname{spt} \varphi \subseteq B_{R+1}$, and

$$\int_{\mathbb{R}^d} |\mathbb{1}_E - \varphi|^p \,\mathrm{d}\mu < \eta.$$

5.c

weight: 0.4 Prove that $C_c(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d, \mu)$.

Free hint.

- In (b), the measure μ need not be finite. Apply regularity to the finite Borel measure $\nu(A) \coloneqq \mu(A \cap B_{R+1})$, and shrink the resulting open set so that it lies inside $\{|x| < R+1\}$.
- In (c), $\|\cdot\|_{L^p}$ is not a norm when $p < 1$. Work with $\|\cdot\|_{L^p}^p$ and the quasi-triangle inequality $\|u+v\|_{L^p}^p \le K_p(\|u\|_{L^p}^p + \|v\|_{L^p}^p)$, $K_p = \max\{1, 2^{p-1}\}$.
- The approximating simple functions of $f\mathbb{1}_{B_R}$ can be chosen to vanish outside B_R.